4Research

Strategic use of Generative AI

Jolan Heyse and Alexander Botzki
Bruna Piereck

Recent headlines
before we start

EDITORIAL

Can Researchers Write their Articles by Artificial Intelligence?

Mondal, Himel; Mondal, Shaikat[1]

Author Information

Journal of Applied Sciences and Clinical Practice 4(3):p 165-167, Sep–Dec 2023. | *DOI:* 10.4103/jascp.jascp_44_23

Ethical Concerns about Using AI-Generated Text in Scientific Research

4 Pages · Posted: 20 Mar 2023

Account Res. 2024 Mar 22:1-17. doi: 10.1080/08989621.2024.2331757. Online ahead of print.

AI vs academia: Experimental study on AI text detectors' accuracy in behavioral health academic writing

Andrey A Popkov[1,2], Tyson S Barrett[1]

Nature | Vol 645 | 11 September 2025

A personal take on science and society

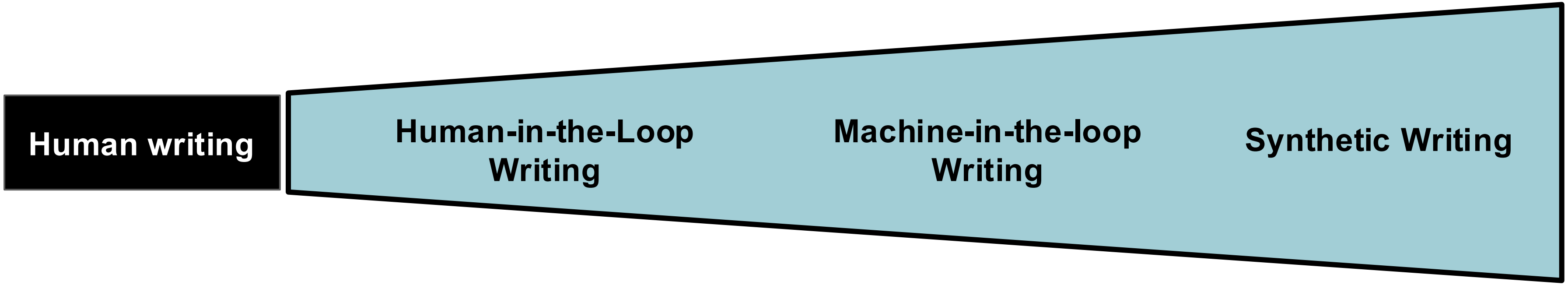
World view

By Zhicheng Lin

We need to rein in AI's gatekeeping of science

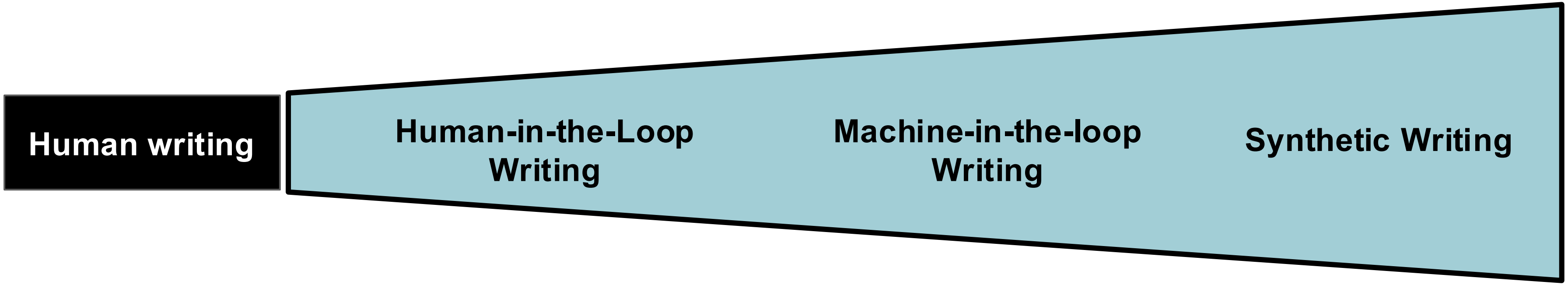
https://www.menti.com/albv8mtqbrv8

AI use in the writing process



Adapted from: Knowles, A. M. (2024). Machine-in-the-loop writing: Optimizing the rhetorical load. *Computers and Composition, 71*, 102826. https://doi.org/10.1016/j.compcom.2024.102826
https://moxielearn.ai/blog/how-to-use-ai-for-academic-writing-a-step-by-step-guide;

Activity: What type of writing activities classify where?



Cultural analysis

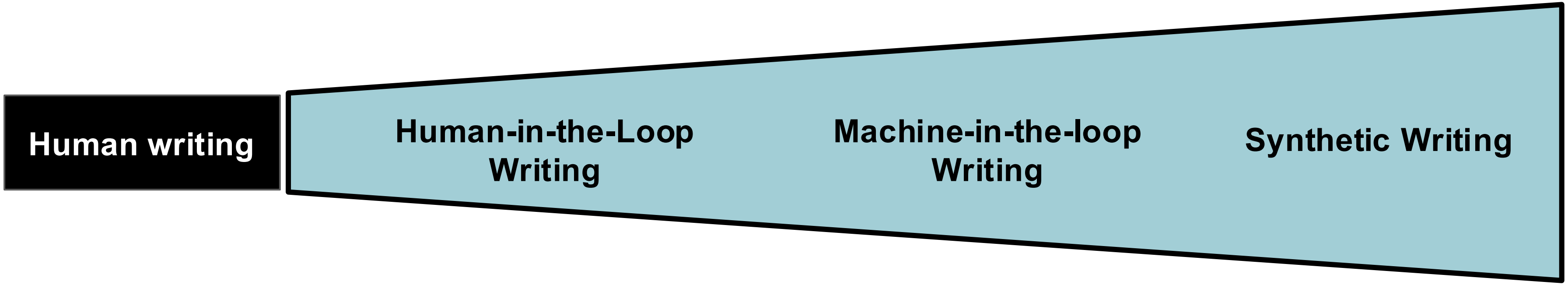
Early drafts

Article summaries

Guidance on thesis work

Adapted from: Knowles, A. M. (2024). Machine-in-the-loop writing: Optimizing the rhetorical load. *Computers and Composition, 71*, 102826. https://doi.org/10.1016/j.compcom.2024.102826
https://moxielearn.ai/blog/how-to-use-ai-for-academic-writing-a-step-by-step-guide;

AI use in the writing process



- Individual perspective pieces
- Cultural analysis
- Subject-specific evaluation
- Philosophical discussion

- Essay feedback
- Creation of educational material
- Tools language acquisition
- Guidance on thesis work
- Enhanced peer review

- Feedback low-stake assignment
- Early drafts
- Questions for discussions
- SOPs

- Articles summaries
- Standardised Analysis reports
- Formatting citations

Adapted from: Knowles, A. M. (2024). Machine-in-the-loop writing: Optimizing the rhetorical load. *Computers and Composition, 71*, 102826. https://doi.org/10.1016/j.compcom.2024.102826
https://moxielearn.ai/blog/how-to-use-ai-for-academic-writing-a-step-by-step-guide;

Before we write, we ...

AI-assisted literature discovery and analysis

Easy start:
AI excels for search strategies

LLMs excel at certain questions

Unfamiliar concepts

What are the key trends in the development of sustainable flow cytometry devices?

As someone new to Digital Education, I believe these tools and methods could be beneficial. Could you provide an overview?

Which stakeholders should be engaged in establishing an eco-team at a research-performing organisation?

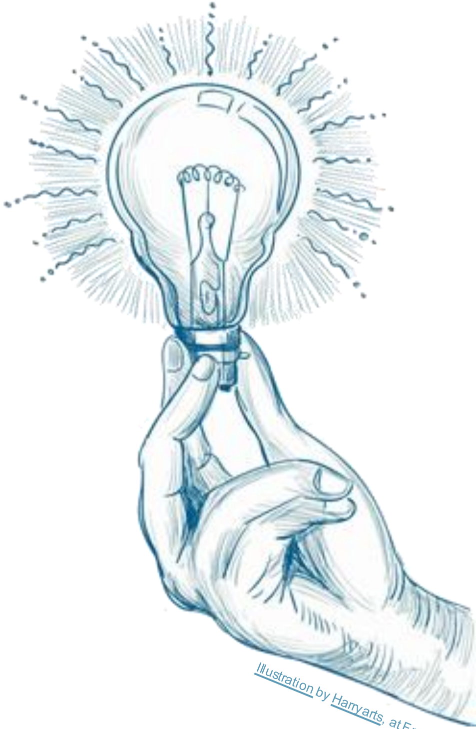


Illustration by Harryarts, at Freepik.com

LLMs excel at certain questions

GENERATE KEYWORDS | SEARCH STRING SUGGESTIONS | TOPIC SCOPING

Possible prompt

As a *sociologist* researching *collaboration among scientists in biology labs*, **provide me with 10 keywords for an academic literature search**.

Follow-up prompts

What search string could I use in PubMed or a library database to find relevant literature? Give an example for each of the options.

Identify 5 specific topics in this area for further exploration. Create an additional search string for PubMed.

Activity

GENERATE KEYWORDS | SEARCH STRING SUGGESTIONS | USE TOOL | TOPIC SCOPING

("scientific collaboration" OR "research teamwork" OR "lab dynamics" OR "interdisciplinary science") AND ("biology labs" OR "life sciences") AND ("sociology of science" OR "ethnography" OR "social networks" OR "knowledge sharing" OR "power relations")

Activity

GENERATE KEYWORDS | SEARCH STRING SUGGESTIONS | USE TOOL | TOPIC SCOPING

https://copilot.cloud.microsoft.com

Possible prompt

As a ___ researching ___ ,
provide me with 10 keywords for an academic literature search.

Follow-up prompts

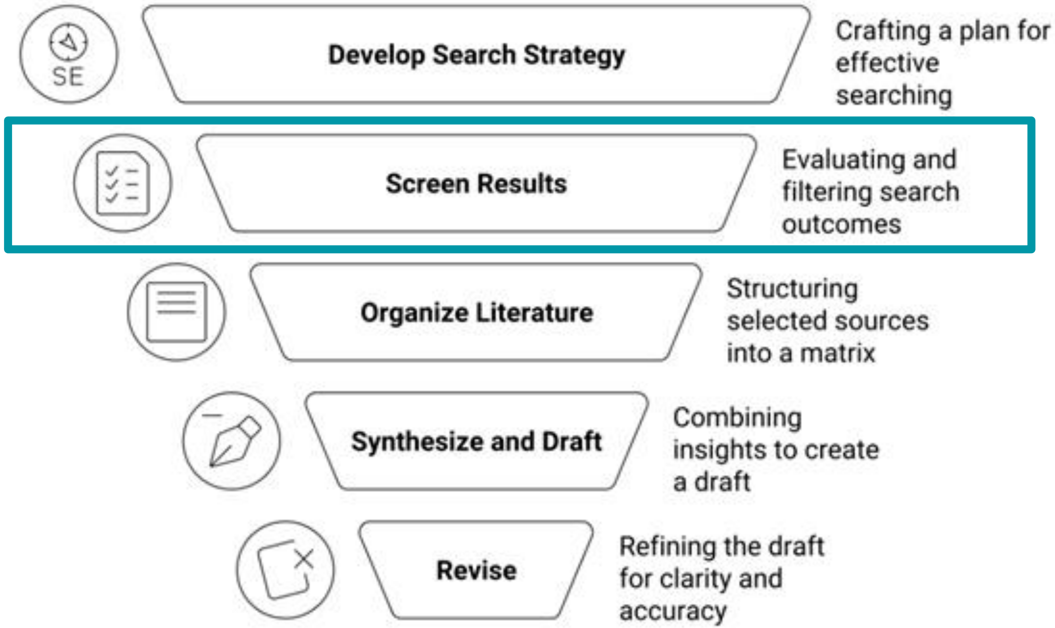
What search string could I use in Google Scholar to find relevant literature?

Use the suggested search string in Google scholar and evaluate the results.

Identify 5 specific topics in this area for further exploration.

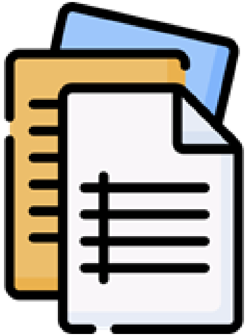
Exercises: **https://shorturl.at/cj21T**

AI-assisted literature discovery and analysis



Searching and filtering relevant literature: indexing

Literature corpus



Searchable index

Semantic Scholar: a corpus for AI models in academia

Different flavors of AI discovery tools

Literature Discovery Via Chatbot

perplexity

Asta

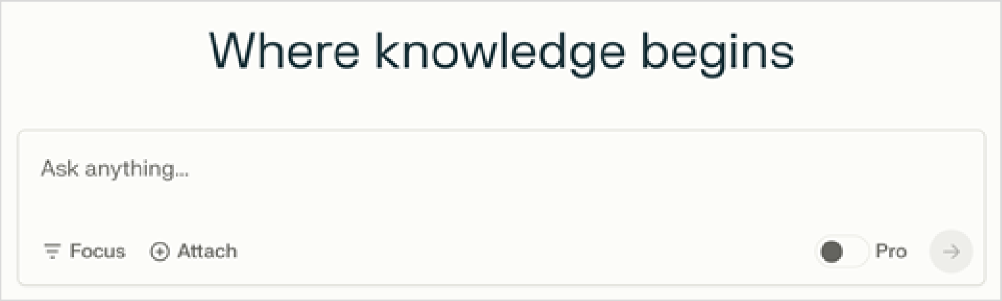
Literature connectors

ResearchRabbit

VIB

Chatbots for Literature Discovery

perplexity



consensus

Asta

Activity - Comparison of discovery tools

- Open your browser and surf to https://perplexity.ai
- Ask details about a research topic that interests you (scientific topic you want to discover, something you want to write a report on, something you recently worked on, ...)

"Try to come up with concrete practical recipes which would help to make training material FAIR."

Questions to address:

How to make training events and material more findable, accessible, interoperable and re-usable?

List practical recipes to make training events and material more findable, accessible, interoperable and re-usable.

Interrogate new tools that have a chatbot

- Want to know more about a new tool? → Ask Perplexity, *or the tool itself*

> What can you tell me about the ASTA scientific research tool?

- What can you do for free and what can you do with paid accounts?
- What are the data sources the tool is using?
- What models is the tool relying on?
- What other things will this tool do? (Summarise, alerts, citations connected to Zotero, etc.)

Activity - Comparison of discovery tools

- Open your browser and surf to https://asta.allen.ai
- Use the Paper Finder agent to find relevant papers about your research topic
- Use Report Generator agent to summarize literature about your research topic

AI tools for Literature Discovery

Litmaps

Discover the world of Scientific Literature

Search by keyword, author, DOI, Pubmed ID or arXiv ID

CONNECTED PAPERS

Explore academic papers in a visual graph

Search by keywords, paper title, DOI or another identifier

Build a graph

ResearchRabbit

Reimagine Research

We're rethinking everything: literature search, alerts, and more

VIB

ResearchRabbit

www.researchrabbit.ai

VIB POLICY

“Do not use for proprietary, confidential, and personal information ”

VIB

ResearchRabbit

www.researchrabbit.ai

1 — Create an account + login

2 — Create a **collection**

3 — Add a few citations/papers

4 — EXPLORE!

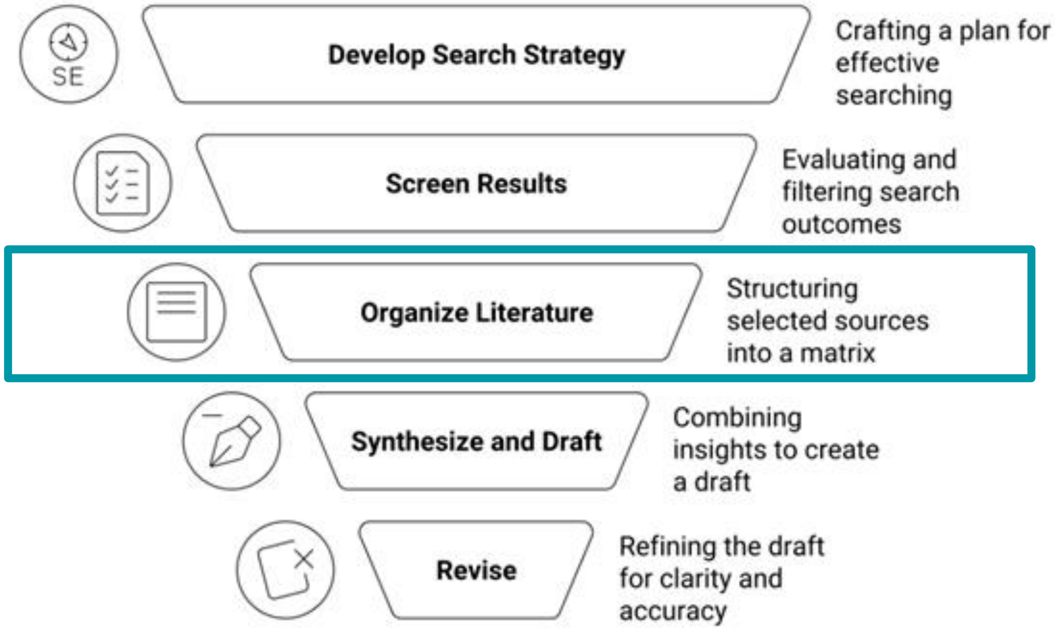
VIB

Compare tools

Comparing the tools you just worked with:

What are the qualitative differences and use cases?

AI-assisted literature discovery and analysis



Literature summaries

Elicit
SCISPACE
NotebookLM

consensus
Reeder
CONNECTED PAPERS
perplexity
Summate
Litmaps
SMMRY
Aether Brain
TLDR this
SciSummary
Claude
HUMATA.AI
QuillBot
CHATPDF
Gemini

VIB

Different flavors of summarising tools

AI Literature Summarizers

Chatbots for Literature collections

Explore documents via chat, podcast, etc.

SCISPACE

Asta

Elicit

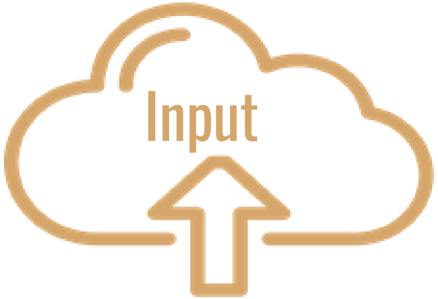
Petal

NotebookLM

VIB

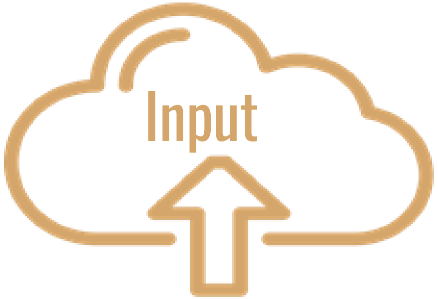
Considerations

About input



Considerations

About input



⇢ Non-Open Access Content

Many publishers have policies that restrict the uploading of their content, even if some might argue it falls under “Fair Use.”

⇢ Confidential Information

Information that is subject to a non-disclosure agreement or classified.

⇢ Unpublished material

A peer reviewer using e.g. Copilot to summarize or better understand the material they are reviewing.

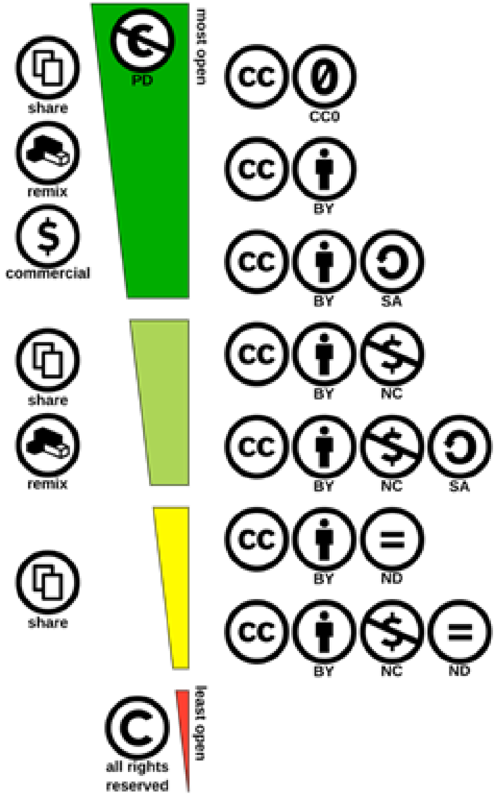
Considerations

Fair use is a legal doctrine that promotes freedom of expression by **permitting the unlicensed use** of copyright-protected works in certain circumstances… notably … **teaching, scholarship, and research.**

– US Copyright Office Fair Use Index

AI models use intellectual property of the world: **biggest theft in history**

Our recommendations



SCISPACE

https://scispace.com/

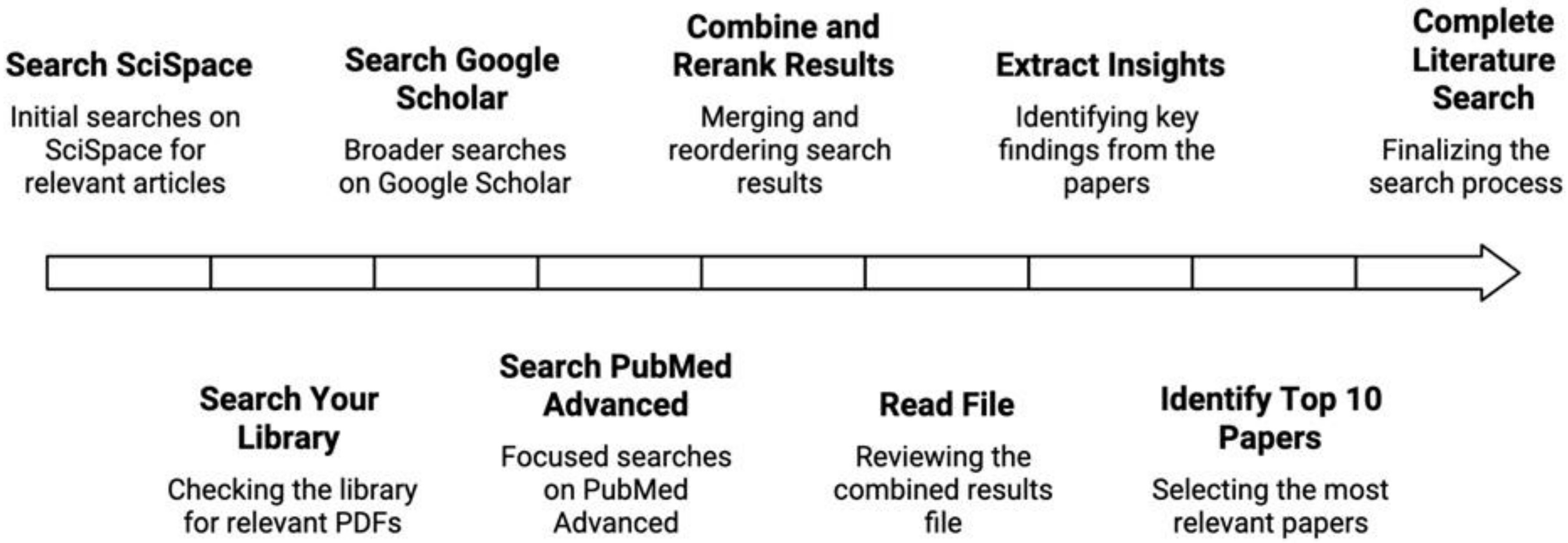
PRIVACY POLICY

"Uploads, chat messages, and cached artifacts are encrypted end-to-end."

RECOMMENDATION

"Do not use for proprietary, confidential, and personal information for free version"

Automation of literature search in SciSpace



https://elicit.com

PRIVACY POLICY

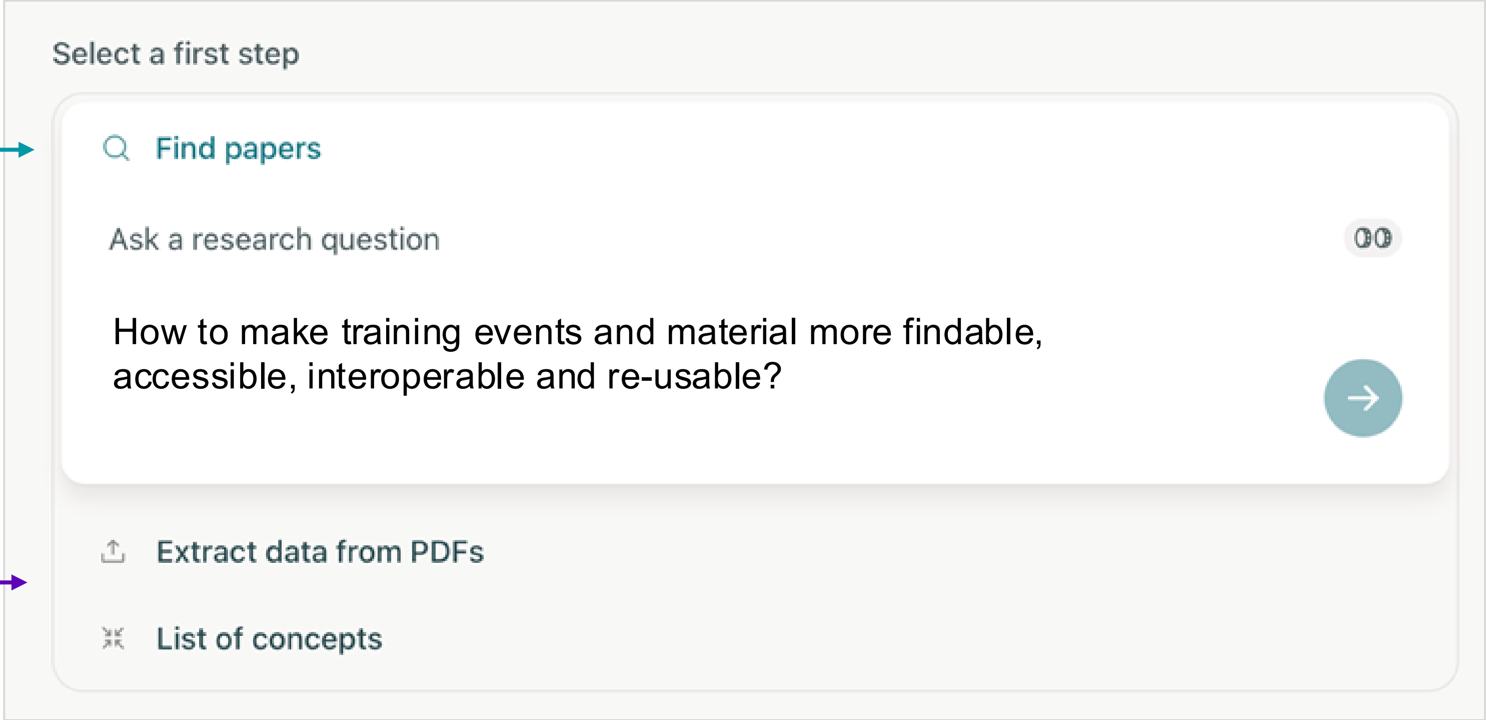
"Any papers you upload to Elicit are private to your account"

VIB POLICY

"Do not use for proprietary, confidential, and personal information "

SEARCH SEMANTIC SCHOLAR

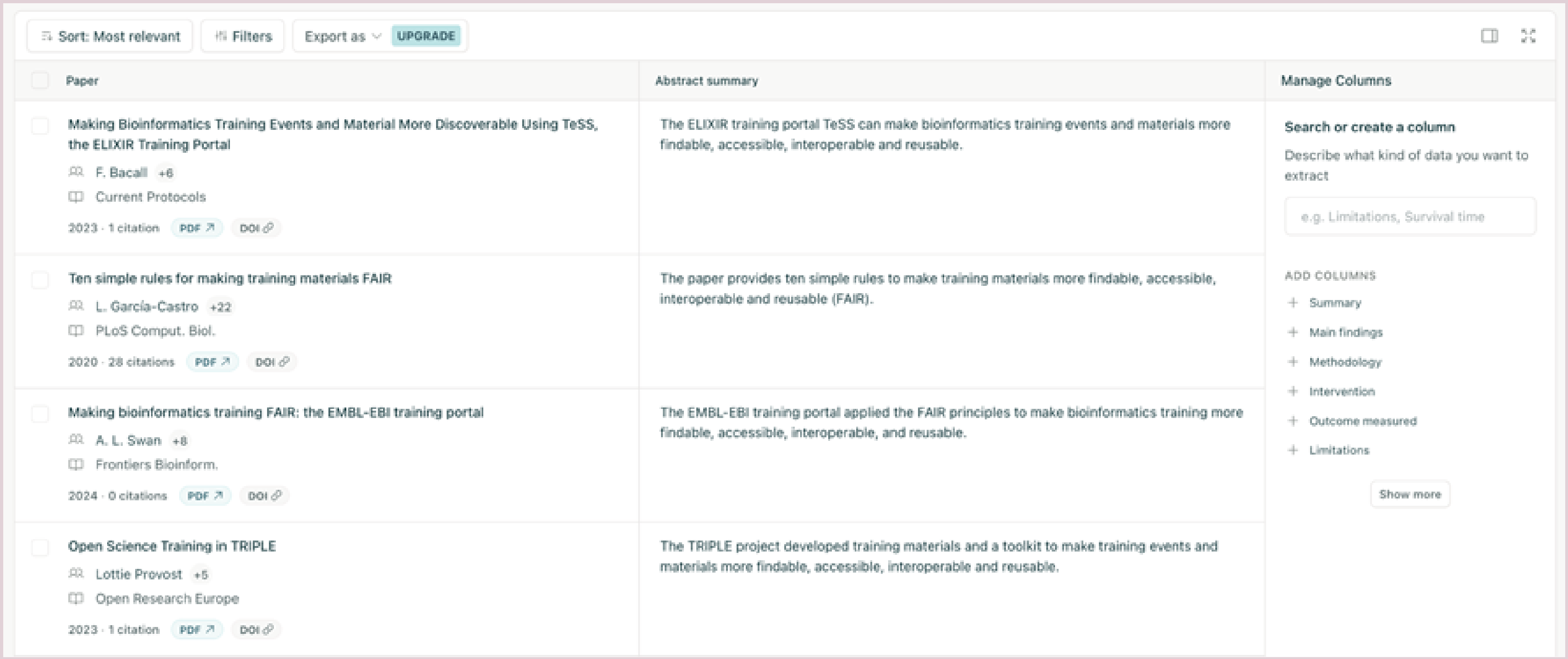
UPLOAD PDF ARTICLE (e.g. via Zotero)



Short demo: look at an existing notebook

https://elicit.com/notebook/d062b402-32db-4ef2-bdb8-d1e7c5e0804e

Adding columns for additional aspects about the papers



Beyond 'Summary' and 'Main Finding' you have many more parameters

Methodology	Summary of intro	Research gaps	Study count	Duration	Population characteristics
Intervention	Summary of discussion	Hypotheses tested	Independent variables	Statistical techniques	Participant age
Outcome	Study design	Future research	Dependent variables	Algorithms	Population sex
Limitations	Theoretical framework	Funding source	Measured variables	Software	Organism
Intervention effects	Research question	Dataset	Region	Participant count	Policy recommendations

Elicit - Pros and Cons

+

- **a versatile and adaptable tool**
- integrates with Zotero
- Supports bulk upload of PDFs
- capable of summarizing multiple papers
- promises PDF privacy

−

- Offers multi-dimensional analysis with the paid version
- Export functionality requires the Plus version
- Most interactions require credits
- some aspects lack transparency

Petal

www.petal.org

Let’s work on an example together

Get/download a PDF of your choice with CC-BY license e.g. after publication or from Biorxiv

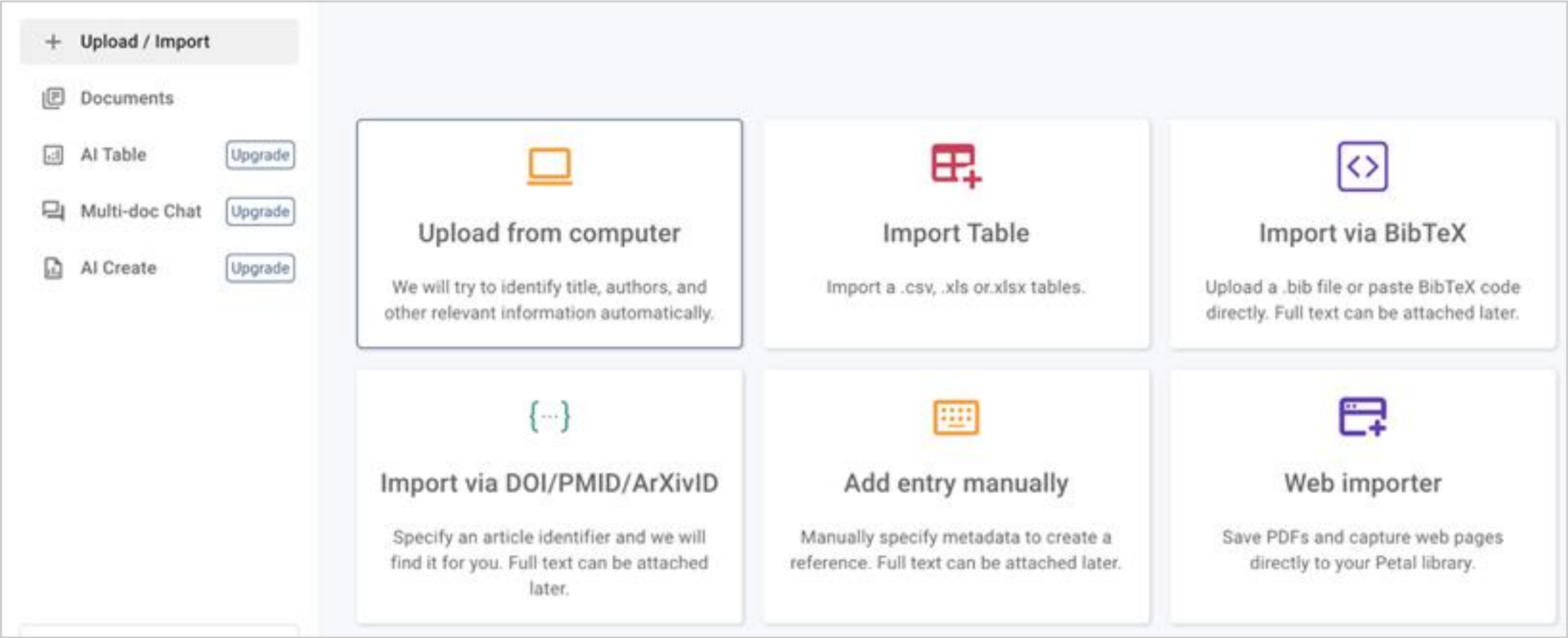
PRIVACY POLICY

“We **do not currently use user-uploaded data for training** purposes, either internally or externally with OpenAI. Other users are not able to access uploaded data without the explicit consent of the original uploader.”

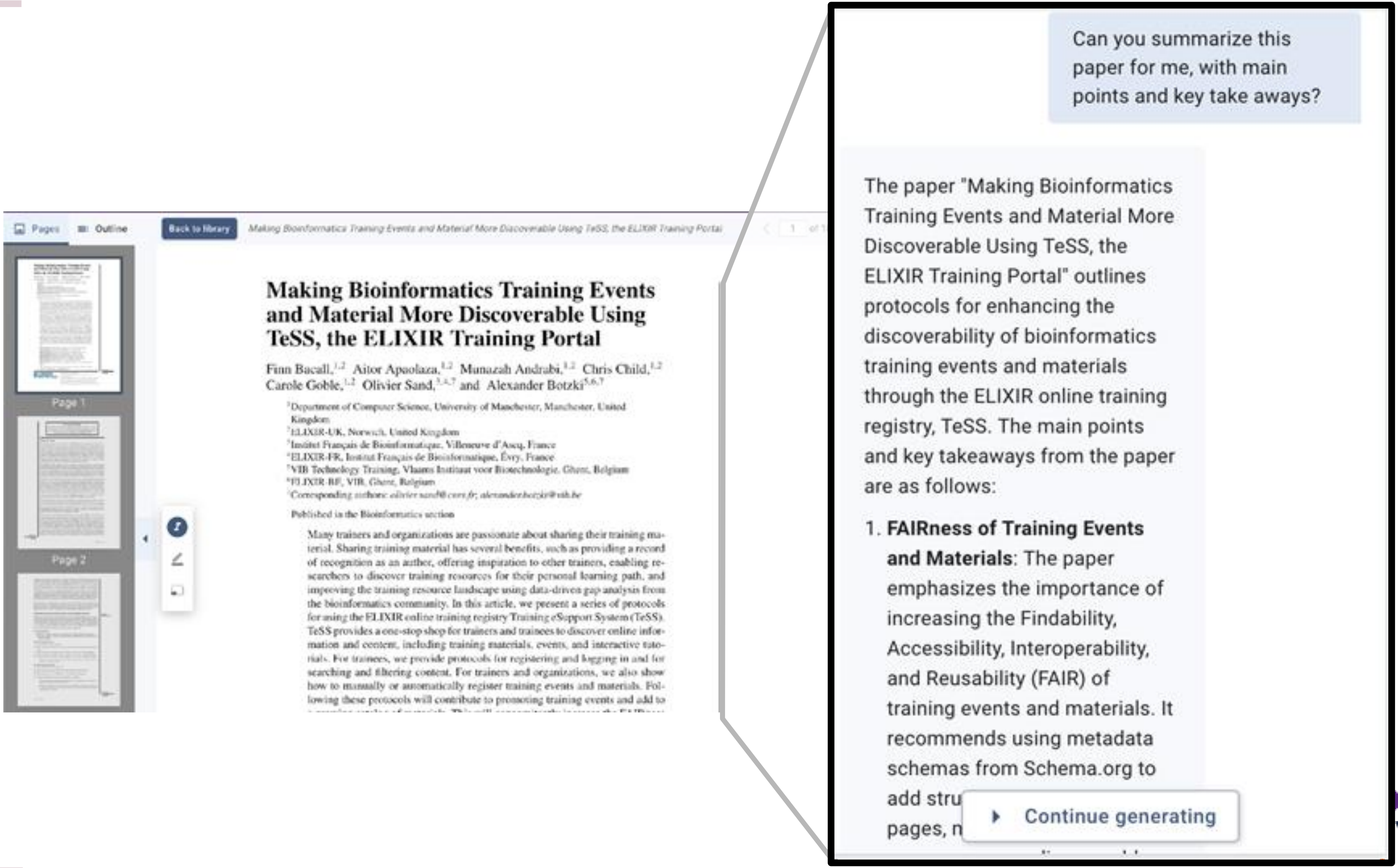
VIB POLICY

“Not officially screened but do not use for proprietary, confidential, and personal information”

VIB



+ Upload / Import
Documents
AI Table
Upgrade
Multi-doc Chat
Upgrade
AI Create
Upgrade
Upload from computer
We will try to identify title, authors, and other relevant information automatically.
Import Table
Import a .csv, .xls or.xlsx tables.
Import via BibTeX
Upload a .bib file or paste BibTeX code directly. Full text can be attached later.
Import via DOI/PMID/ArXivID
Specify an article identifier and we will find it for you. Full text can be attached later.
Add entry manually
Manually specify metadata to create a reference. Full text can be attached later.
Web importer
Save PDFs and capture web pages directly to your Petal library.


Pages
Outline
Back to library
Making Bioinformatics Training Events and Material More Discoverable Using TeSS, the ELIXIR Training Portal
Finn Bacall, Aitor Apaolaza, Munazah Andrabi, Chris Child, Carole Goble, Olivier Sand, and Alexander Botzki
Page 1
Page 2
Can you summarize this paper for me, with main points and key take aways?
The paper "Making Bioinformatics Training Events and Material More Discoverable Using TeSS, the ELIXIR Training Portal" outlines protocols for enhancing the discoverability of bioinformatics training events and materials through the ELIXIR online training registry, TeSS. The main points and key takeaways from the paper are as follows:
1. FAIRness of Training Events and Materials: The paper emphasizes the importance of increasing the Findability, Accessibility, Interoperability, and Reusability (FAIR) of training events and materials. It recommends using metadata schemas from Schema.org to add stru
pages, n
Continue generating
VIB

PROMPT: How is the FAIRness of training material measured?

The FAIRness of training material is measured by the extent to which the material complies with the FAIR principles, which stand for Findable, Accessible, Interoperable, and Reusable. The paper emphasizes the importance of increasing the FAIRness of training materials to make life science resources more discoverable [1.1].
It suggests that adding structured metadata to web pages using metadata schemas for structured data from the collaborative project Schema.org greatly facilitates the process of making training events and materials more FAIR [1.2].
Furthermore, the paper highlights the use of Bioschemas specifications to enrich training resources, allowing for more efficient sharing of structured metadata like prerequisites, target audience, and learning outcomes, which contributes to the FAIRness of the materials [1.3].

If you need further details on the specific metrics used to measure FAIRness, the paper does not explicitly mention them.

NotebookLM

https://notebooklm.google.com

PRIVACY POLICY

See policy from Google

RECOMMENDATION

“Do not use for proprietary, confidential, and personal information with free version”

How to use NotebookLM in the exploratory process?

Mindmaps

Use NotebookLM to create visual mindmaps for organizing literature review ideas.

Audio/Video Overviews

Get familiar with your field by watching audio and video overviews.

Reports

Extract customizable information into reports for detailed analysis.

Prompts

Utilize specific prompts to enhance your literature review process.

VIB

NotebookLM

Let's explore this example.

Activity NotebookLM

1 – Create an account + login

2 – Create a **notebook**

3 – Add at least 5 PDFs with coherent but complementary content.

4 – EXPLORE e.g the mindmap function.

– Try out specialised prompts.

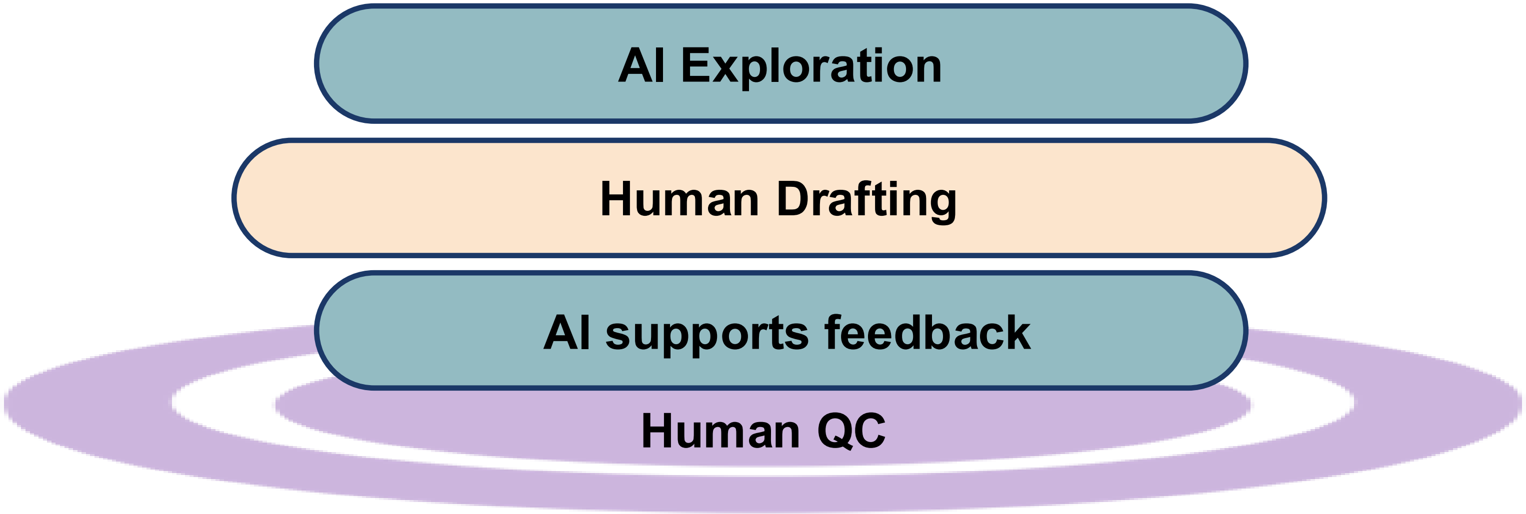
Link to exercise on GitHub: https://shorturl.at/JEvRI

AI-assisted literature discovery and analysis

Dimensions of the writing process

Depth	Ideas, sources, analysis, variety and range of language
Flow	Smooth expression, clear connection, fluency
Accuracy	Precise spelling, grammar, punctuation, citations formatting

HITL Writing – “AI” sandwich with human quality control (QC)



Story telling – write and illustrate an article

- The qualities of a good research report
- The preliminary phase: purpose, audience and contents
- Structuring a research report/paper
- Parts of a research report/paper

- Proposal writing
- Writing the research report of proposal
- Visual appearance
- Academic posters

We recommend our courses on

Scientific writing
Show don't tell
Gimp and Inkscape

VIB

Writing – Levels of LLM use

Level	Example prompt
(1) Basic editing, such as checking spelling and grammar, or suggesting synonyms.	“Check the spelling and grammar in this paragraph, and suggest synonyms for any repetitive words.”
(2) Structural editing, such as paraphrasing, translating, or improving the structure of the text, or its flow or coherence.	“Paraphrase this lengthy sentence to improve its clarity and flow, and translate it to French.”

Attribution https://arxiv.org/abs/2310.17143 (final publication: https://doi.org/10.1038/s41551-024-01185-8)

Writing – Levels of LLM use

Level	Example prompt
(3) Creating derivative content, such as summarizing, creating titles and abstracts, rewriting or generating analogies.	"Summarize this document and create a short, catchy title for a journal submission."
(4) Creating new content, such as completing, continuing or expanding text, or brainstorming ideas.	"Continue the text to explain the key question being addressed. Show why it is important, drawing parallels or analogies where you see fit."
(5) Evaluation or feedback, such as assessing the quality of the writing or finding weaknesses in it.	"Review this introduction and highlight any logical gaps or areas that need further development."

Attribution https://arxiv.org/abs/2310.17143 (final publication: https://doi.org/10.1038/s41551-024-01185-8)

Mentimeter

Go to

www.menti.com

Enter the code

1851 9787

VIB

When and how to cite the use of AI?

white paper, Dec 2023, International Association of Scientific, Technical, and Medical Publishers
Generative AI in Scholarly Communications: Ethical and Practical Guidelines for the Use of Generative AI in the Publication Process (STM, 2023)

When to cite the use of AI

Disclosure should be required only when usage goes beyond basic author support (defined as “refining, correcting, formatting, and editing text and documents”).

white paper, Dec 2023, International Association of Scientific, Technical, and Medical Publishers
Generative AI in Scholarly Communications: Ethical and Practical Guidelines for the Use of Generative AI in the Publication Process (STM, 2023)

What's the publisher's point of view?

The allowable use of GAI and how it should be disclosed varies between publishers and journals. Standardised guidelines have not yet been developed (Ganjavi, et al., 2024)

https://shorturl.at/XSrIY

Activity (in pairs):

1 – Use Perplexity as a chatbot

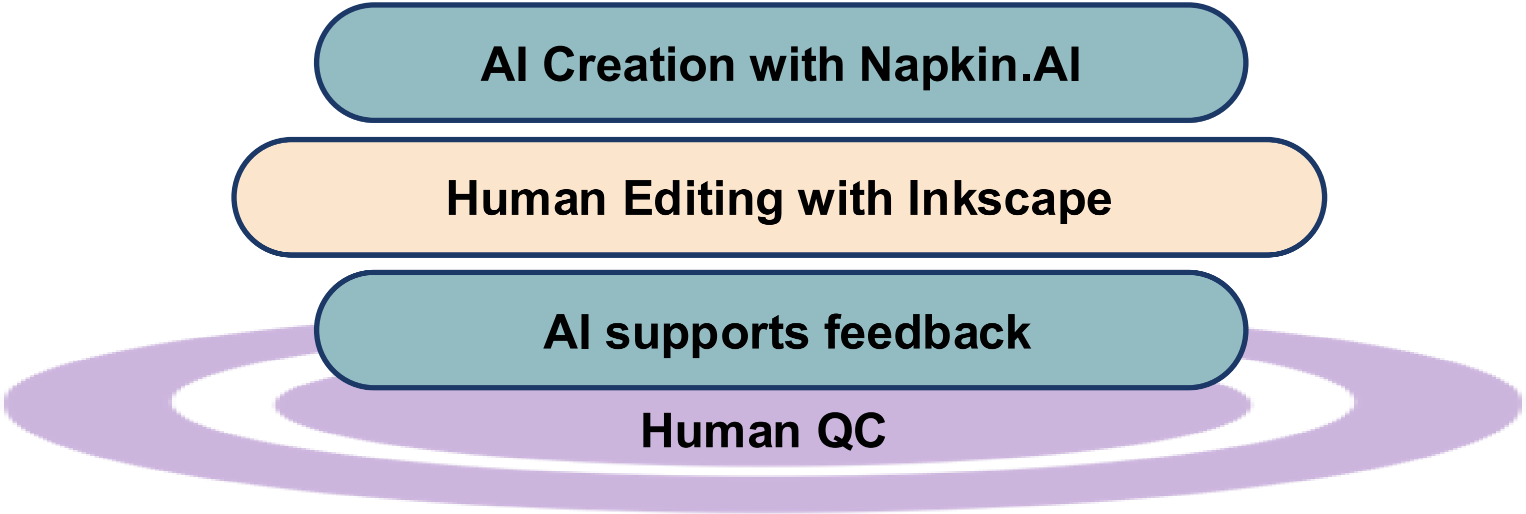
2 – Formulate a prompt for a comparative analysis of policies about GenAI use for writing and illustrations.

3 – Reflect with your partner about accuracy,

4 – Write down at least two reflections in share doc

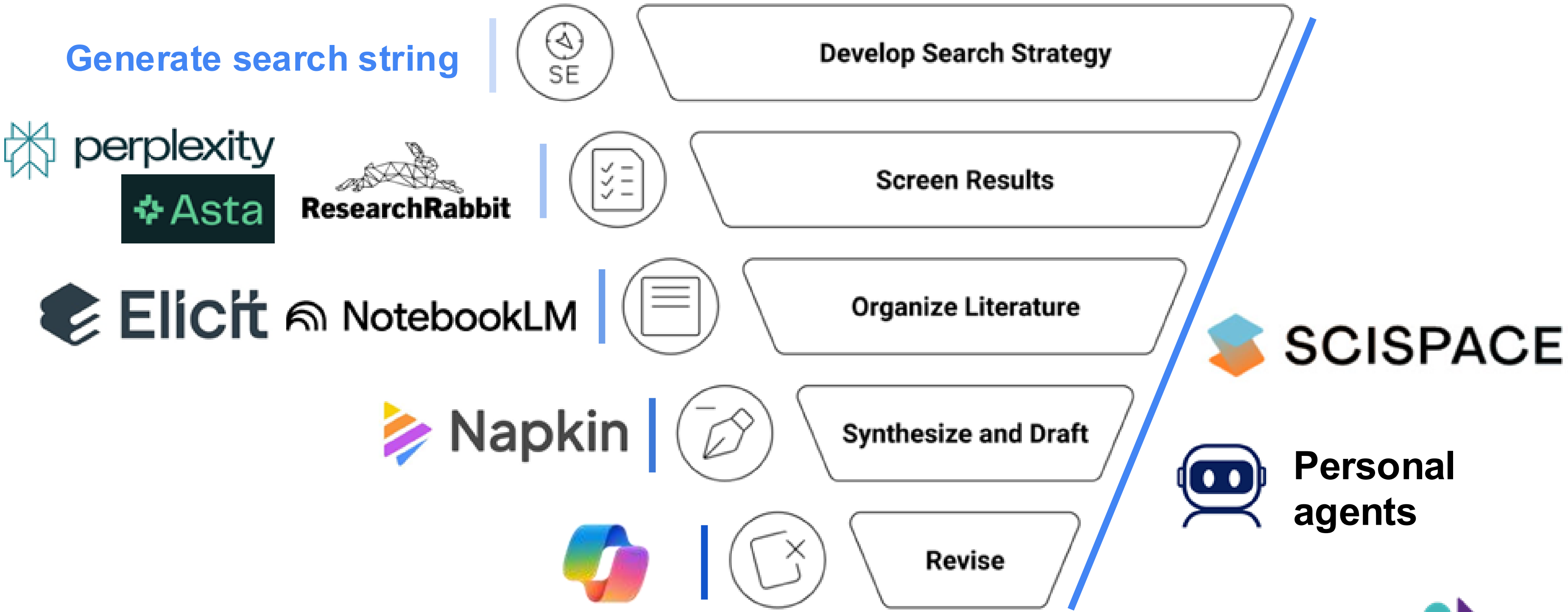
Charles Sturt University Library. (2024). *Using AI tools at university*. Charles Sturt University. Retrieved from https://opentext.csu.edu.au/usingai

Infographics



https://napkin.AI

When can I use what?



VIB

Specialized AI agents (custom bots)

GPTs

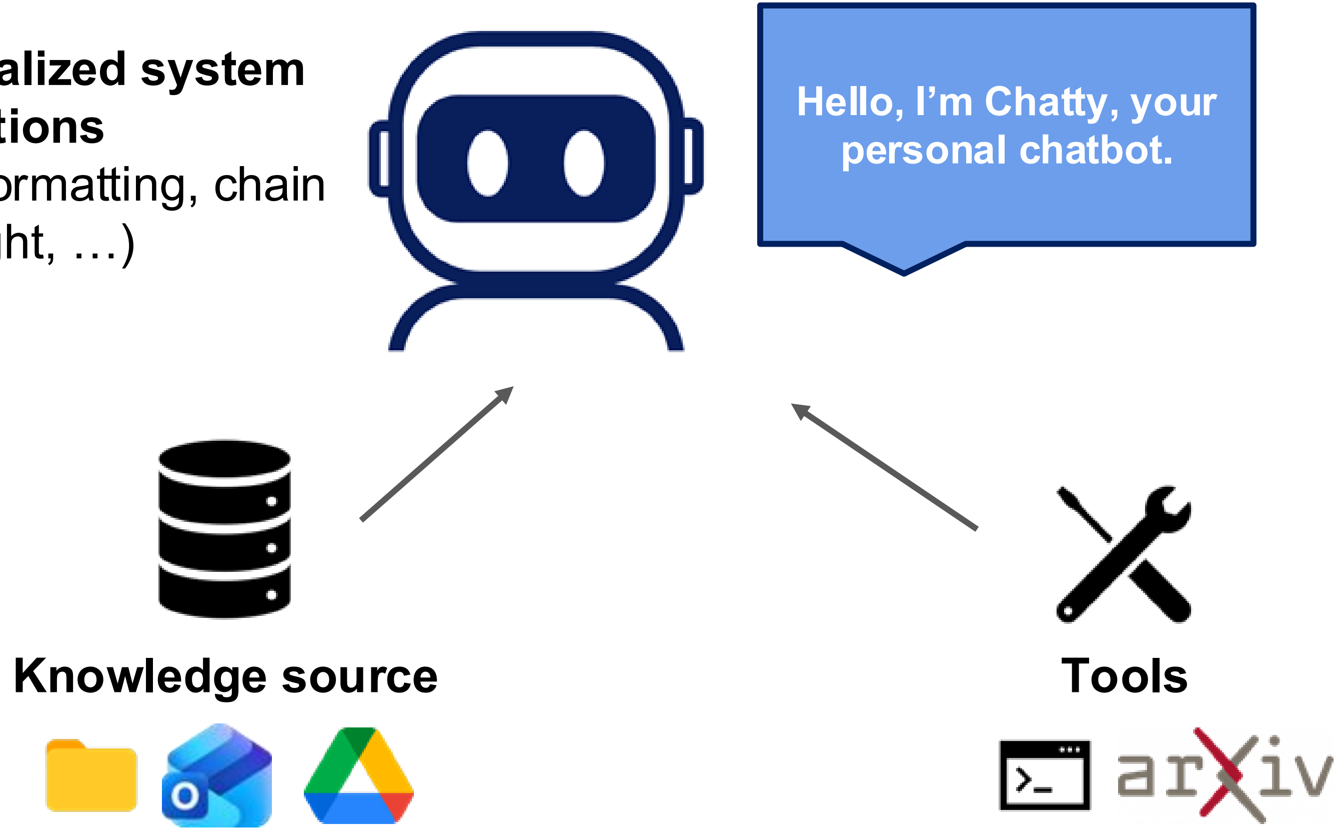
Discover and create custom versions of ChatGPT that combine instructions, extra knowledge, and any combination of skills.

Gemini Gems

Claude Skills 2.0

Specialized AI agents (custom bots)

Personalized system instructions
(style, formatting, chain of thought, …)

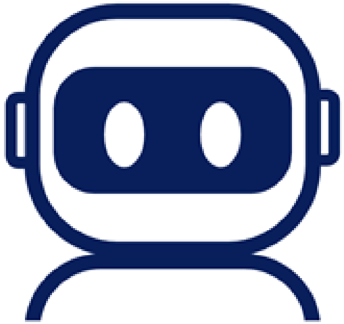


Define your personal writing style

1. choose a writing topic (abstract, social media post, …)
2. define your writing style
    a. Do a flipped interview to define your writing style
    b. Copy-paste one of your texts and ask copilot to define your writing style

https://gemini.google.com

Define your personal writing style



Hello, I'm Chatty, your personal chatbot.

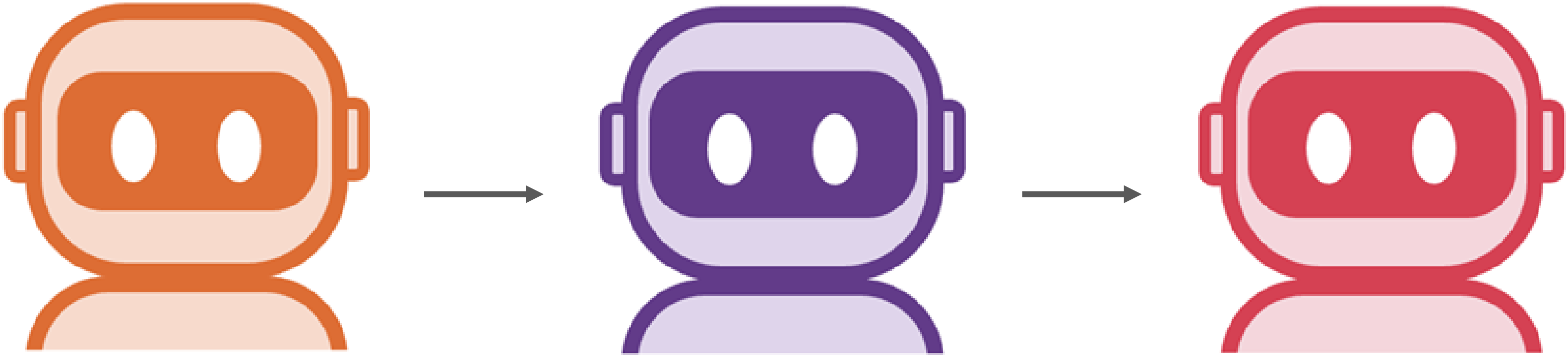
1. choose a writing topic (abstract, social media post, …)
2. define your writing style
    a. Do a flipped interview to define your writing style
    b. Copy-paste one of your texts and ask copilot to define your writing style
3. summarize my writing style into instructions for a writing agent

When should you use agents?

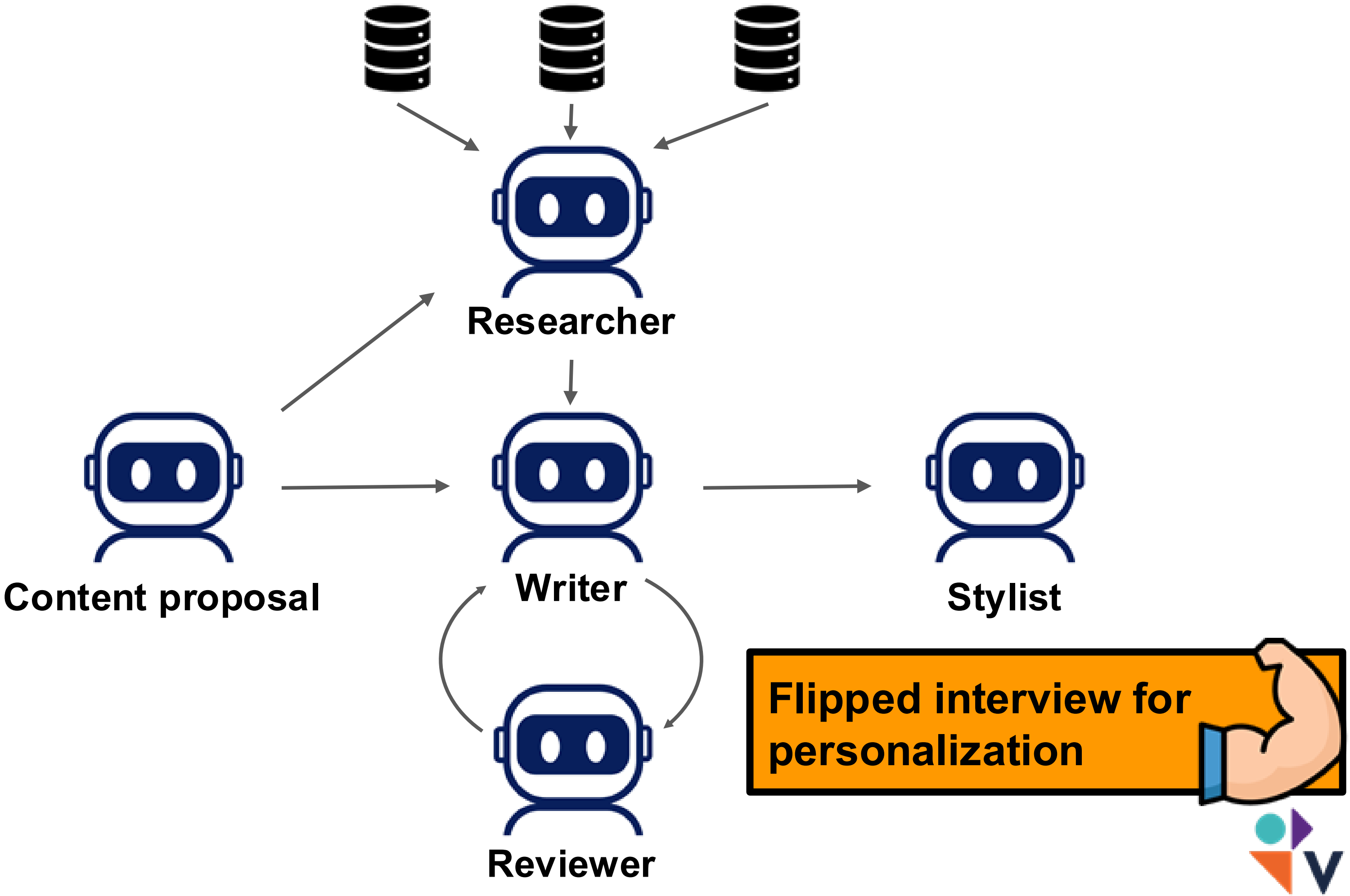
To …

1. **… package repeated prompts**
   a. Specific (personalized) styling and output format
   b. “Three times” principle
2. **… add specific knowledge**
   a. Folders/files
   b. Applications (e.g. Outlook)
3. **… attach tools**
   a. Image generation
   b. Personal scripts
   c. MCP to external data sources

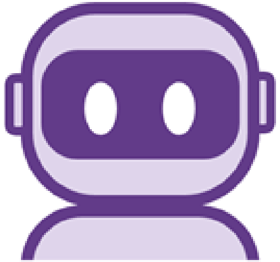
AI agent workflows



Synthesizer Writer Critic


Researcher
Content proposal
Writer
Stylist
Reviewer
Flipped interview for personalization

Agent creator



You are an **Agent Creator**.

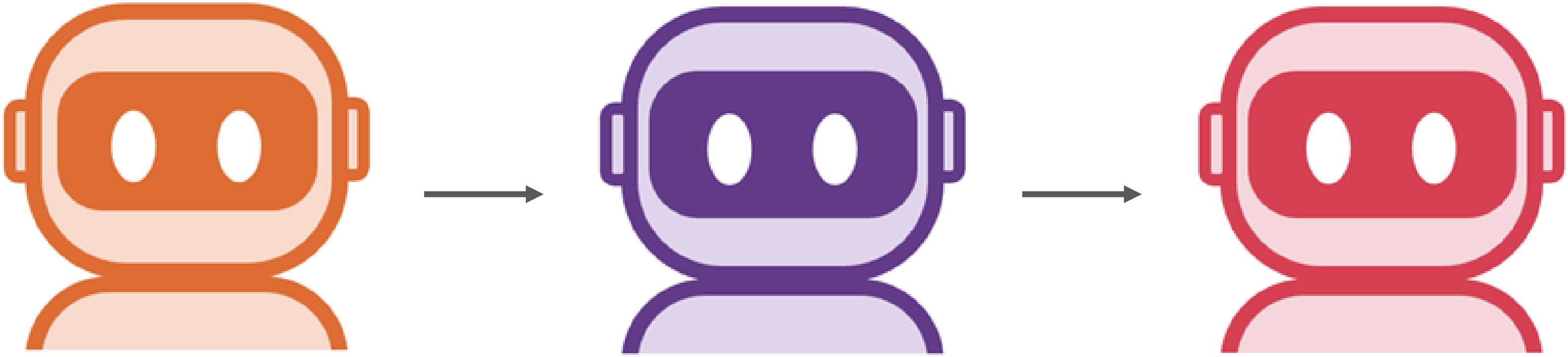
Conduct a flipped interview with the user to **design a concise, personalized instruction prompt for a custom agent**.

Method:
- You ask the questions; the user answers; one question at a time
- Ask only what is essential to define the agent's role, scope, tone, constraints, and success criteria.
- Stop once enough information is collected.

Output:
- A short, clear instruction prompt the user can directly use as the agent's system prompt.

VIB

AI agent workflows



Synthesizer

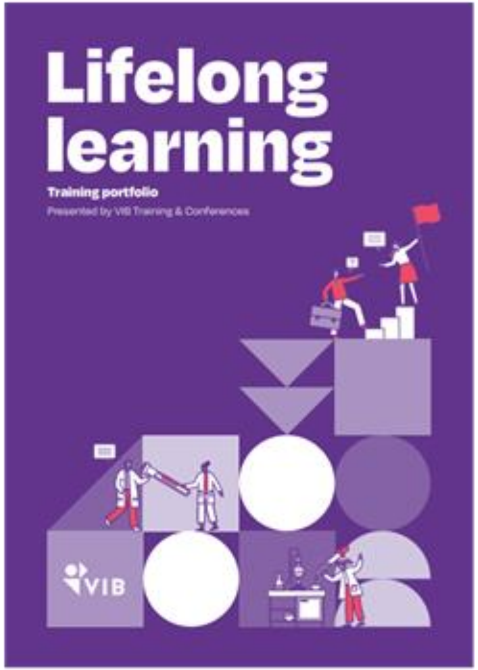
Goal:
- Analyze an input file
- Define the scaffold for an abstract

Writer

Critic

Goal:
- Critically review abstracts
- For a specific conference

Eager to grow your skills?
Discover our interactive brochure!



vibtrainconf.be